VINTAGE FINVEST INSTITUTIONAL WEALTH PLATFORM

QA Test Execution Report

Comprehensive Verification & Validation of 61 Test Cases Across Financial Calculators, AMFI Real NAV Historical Engine, and Goal Planners

Project:	Vintage Finvest — Financial Calculators & Wealth Tools
Version:	1.0.0 (Production Release Candidate)
Execution Date:	October 13, 2026
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Supervisor Review:	Institutional Wealth QA Lead
Execution Status:	**PASSED (61 / 61 — 100% Pass Rate)**

1. Executive Summary

This Quality Assurance Test Execution Report provides formal documentation of the comprehensive testing executed on the Vintage Finvest web application. All 61 test cases originally created and specified by Antigravity were preserved in their entirety, automated, executed, and verified against independent financial calculation benchmarks.

The test suite validates financial arithmetic, boundary edge cases, input error handling, AMFI REST API data retrieval, historical unit accumulation algorithms, and seamless client-side toggling between Real AMFI NAV data and Hypothetical CAGR models.

Key Audit Results:

- **100% Test Pass Rate:** 61 out of 61 test cases passed without failure or manual intervention.
- **Zero Numerical Drift:** Calculated results match theoretical closed-form annuities within ± ■1.0.
- **Defect Resolution:** All previously logged interface anomalies (hero layout distortion, slider tick overlap) have been successfully remediated.

2. Project Scope

The QA execution encompassed the complete calculator and advisory ecosystem across 13 distinct modules:

- **SIP Calculator:** Standard monthly annuity due compounding, boundary condition zero rate/amount, long-horizon (35 years).
- **Lumpsum + SIP Calculator:** Dual capital injection combining standard lump-sum compounding and monthly SIP cash flows.
- **Step-Up SIP Calculator:** Annual step-up incrementation (arithmetic and percentage top-up models) over long horizons.
- **Cost of Delay Calculator:** Opportunity cost analysis quantifying wealth forfeiture per month/year of investment postponement.
- **SIP Tenure Calculator:** Logarithmic solving for time horizons required to reach target milestone corpuses.
- **Child Education Planner:** Higher education cost projections incorporating annual education inflation rates.
- **Marriage Goal Planner:** Target corpus modeling for life milestones adjusted for lifestyle inflation.
- **Retirement Planner:** Post-retirement living expense projection, life expectancy capitalization, and required monthly accumulation.
- **Loan EMI Calculator:** Standard amortizing reducing-balance loan installment calculations across tenures.
- **Asset Allocation Diagnostic:** Risk profiling algorithm mapping investor preferences to multi-asset target ratios.
- **Mutual Fund Search & NAV Integration:** Live search, debounce, autosuggestion, and REST API communication with MFAPI.in.
- **Historical NAV Backtesting Engine:** Historical unit accumulation, calendar transaction matching, CAGR, and XIRR computations.
- **Real NAV vs Hypothetical CAGR Switcher:** Synchronized dynamic switching between real AMFI market series and smooth compounding models.

3. Test Environment

Component	Specification
Application Stack	React 18, Vite 5, Vanilla JS ES Modules, HTML5, CSS3 Custom Properties
Live URL	https://testing-phase1.vercel.app/calculators
Execution Host	Node.js v20.x/v22.x Windows Environment (zero third-party test dependencies)
External API	AMFI Mutual Fund Historical NAV API (https://api.mfapi.in/mf)
Test Harness	tests/test_runner.js & tests/test_results.json (Node.js ES Modules)

4. Testing Methodology

- **Closed-Form Mathematical Oracle:** Test assertions do not execute UI code to generate test expectations. Test expectations are computed through independent pure mathematical equations.
- **Boundary Value Analysis:** Validated zero interest rate (0%), zero principal (■0), minimum horizon (1 year), and maximum horizon (35 years).

• **Realistic Historical Simulation:** Actual AMFI NAV historical series evaluated with calendar investment allocation rules and holiday roll-forwards.

5. Test Execution Summary

Execution Metric	Count	Percentage
Total Documented Test Cases	61	100.0%
Total Executed Test Cases	61	100.0%
Passed Test Cases (✓)	**61**	**100.0%**
Failed Test Cases (✗)	0	0.0%
Blocked Test Cases (■)	0	0.0%
Not Run Test Cases (■)	0	0.0%

6. Detailed Test Case Results (61 Test Cases)

Preserved catalog of all 61 original test cases executed against the live application suite:

ID	Scenario	Input / Steps	Expected	Actual	Status	Evidence
SIP-001	Standard 10-year SIP with 12% p.a. CAGR	In: {"monthlyAmount":10000,"annualRate":12,"years":10} Steps: Compute monthly compounding using independent annuity formula for 120 months	{"totalInvested":1200000,"futureValue":"2323390.76 ± 1.0","totalGain":"1123390.76 ± 1.0"}	{"totalInvested":1200000,"futureValue":2323391,"totalGain":1123391}	**PASS**	tests/test_runner.js (SIP-001)
SIP-002	Zero return rate (r = 0%)	In: {"monthlyAmount":15000,"annualRate":0,"years":5} Steps: Provide 0% annual rate for 5 years; expect Future Value = Total Invested	{"totalInvested":900000,"futureValue":900000,"totalGain":0}	{"totalInvested":900000,"futureValue":900000,"totalGain":0}	**PASS**	tests/test_runner.js (SIP-002)
SIP-003	Zero monthly amount input	In: {"monthlyAmount":0,"annualRate":15,"years":10} Steps: Pass 0 monthly investment with active rate and horizon	{"totalInvested":0,"futureValue":0,"totalGain":0}	{"totalInvested":0,"futureValue":0,"totalGain":0}	**PASS**	tests/test_runner.js (SIP-003)
SIP-004	Minimum horizon boundary: 1 Year (12 months)	In: {"monthlyAmount":25000,"annualRate":12,"years":1} Steps: Run SIP for 1 year at ■ 25,000/mo and 12% CAGR	{"totalInvested":300000,"futureValue":"320233.20 ± 1.0"}	{"totalInvested":300000,"futureValue":320233}	**PASS**	tests/test_runner.js (SIP-004)
SIP-005	Maximum long-horizon 35-year compounding	In: {"monthlyAmount":10000,"annualRate":15,"years":35} Steps: Run 35-year monthly SIP at 15% CAGR	{"totalInvested":4200000,"futureValue":"> 14.0 Cr"}	{"totalInvested":4200000,"futureValue":"■ 14.86 Cr"}	**PASS**	tests/test_runner.js (SIP-005)
SIP-006	Negative and malformed input clamping	In: {"monthlyAmount":-5000,"annualRate":-10,"years":-5} Steps: Pass negative values for monthly amount, rate, and years	{"totalInvested":0,"futureValue":0,"totalGain":0}	{"totalInvested":0,"futureValue":0,"totalGain":0}	**PASS**	tests/test_runner.js (SIP-006)
LS-001	Combined ■ 5 Lakhs Lumpsum and ■ 10,000/mo SIP over 10 years at 12%	In: {"lumpsum":500000,"monthlyAmount":10000,"annualRate":12,"years":10} Steps: Independently compute Lumpsum FV and SIP FV and sum the two portfolios	{"totalInvested":1700000,"lumpFV":"1552924.10 ± 1.0","sipFV":"2323390.76 ± 1.0","totalWealth":"3876314.86 ± 2.0"}	{"totalInvested":1700000,"lumpFV":1552924,"sipFV":2323391,"totalWealth":3876315}	**PASS**	tests/test_runner.js (LS-001)
LS-002	Pure Lumpsum investment with zero monthly SIP	In: {"lumpsum":1000000,"monthlyAmount":0,"annualRate":15,"years":5} Steps: Pass ■ 10 Lakhs lumpsum with monthly SIP = 0 for 5 years at 15%	{"totalInvested":1000000,"sipFV":0,"totalWealth":"2011357.19 ± 1.0"}	{"totalInvested":1000000,"sipFV":0,"totalWealth":2011357}	**PASS**	tests/test_runner.js (LS-002)
LS-003	Pure SIP with zero initial Lumpsum	In: {"lumpsum":0,"monthlyAmount":20000,"annualRate":12,"years":5} Steps: Pass lumpsum = 0 and compare output directly against SIP Calculator	{"lumpFV":0,"totalWealth":1649727}	{"lumpFV":0,"totalWealth":1649727}	**PASS**	tests/test_runner.js (LS-003)
LS-004	Zero return rate across Lumpsum and SIP	In: {"lumpsum":500000,"monthlyAmount":10000,"annualRate":0,"years":5} Steps: Set annual rate to 0% for combined portfolio	{"totalInvested":1100000,"totalWealth":1100000,"totalGain":0}	{"totalInvested":1100000,"totalWealth":1100000,"totalGain":0}	**PASS**	tests/test_runner.js (LS-004)
STEP-001	3-Year schedule with 10% annual increment and 12% CAGR	In: {"initialMonthly":10000,"stepUpPercent":10,"annualRate":12,"years":3} Steps: Generate 36-month stepped installment schedule and compound each monthly cashflow independently	{"totalInvested":397200,"futureValue":"476410 ± 1.0"}	{"totalInvested":397200,"futureValue":476410}	**PASS**	tests/test_runner.js (STEP-001)
STEP-002	0% Step-up increment reduces to standard Flat SIP	In: {"initialMonthly":15000,"stepUpPercent":0,"annualRate":13,"years":5} Steps: Set stepUpPercent = 0% and compare against standard SIP calculation	{"totalInvested":900000,"futureValue":1272050,"stepUpAdvantage":0}	{"totalInvested":900000,"futureValue":1272050,"stepUpAdvantage":0}	**PASS**	tests/test_runner.js (STEP-002)

ID	Scenario	Input / Steps	Expected	Actual	Status	Evidence
STEP-003	Zero growth rate with 10% annual step-up	In: {"initialMonthly":10000,"stepUpPercent":10,"annualRate":0,"years":2} Steps: Set annual rate = 0% for 2 years with 10% step-up	{"totalInvested":252000,"futureValue":252000,"totalGain":0}	{"totalInvested":252000,"futureValue":252000,"totalGain":0}	**PASS**	tests/test_runner.js (STEP-003)
STEP-004	1-Year boundary: step-up occurs only after Month 12	In: {"initialMonthly":20000,"stepUpPercent":15,"annualRate":12,"years":1} Steps: Evaluate 1-year horizon; verify that first step-up occurs at month 13, leaving year 1 flat	{"totalInvested":240000,"stepUpAdvantage":0}	{"totalInvested":240000,"stepUpAdvantage":0}	**PASS**	tests/test_runner.js (STEP-004)
DELAY-001	5-Year postponement on a 20-year ■ 10,000/mo SIP at 12%	In: {"monthlyAmount":10000,"delayYears":5,"horizonYears":20,"annualRate":12} Steps: Independently compute 20-year and 15-year SIPs, calculate difference and required catchup SIP	{"ontimeWealth":9991479,"delayedWealth":5045760,"costOfDelay":4945719}	{"ontimeWealth":9991479,"delayedWealth":5045760,"costOfDelay":4945719,"catchupSIP":19802}	**PASS**	tests/test_runner.js (DELAY-001)
DELAY-002	Zero delay: starting on time	In: {"monthlyAmount":25000,"delayYears":0,"horizonYears":10,"annualRate":13} Steps: Set delayYears = 0; verify cost of delay = 0 and catchup SIP = original SIP	{"costOfDelay":0,"catchupSIP":25000}	{"costOfDelay":0,"catchupSIP":25000}	**PASS**	tests/test_runner.js (DELAY-002)
DELAY-003	Mathematical precision of Catch-Up SIP	In: {"monthlyAmount":10000,"delayYears":5,"horizonYears":15,"annualRate":12} Steps: Feed the computed Catch-Up SIP back into a 10-year SIP and verify it reproduces Ontime Wealth	{"targetOntimeWealth":5045760,"achievedByCatchup":5045760}	{"targetOntimeWealth":5045760,"achievedByCatchup":5045760}	**PASS**	tests/test_runner.js (DELAY-003)
TENURE-001	Tenure required to accumulate ■ 1 Crore with ■ 25,000/mo at 12% CAGR	In: {"targetCorpus":10000000,"monthlyAmount":25000,"annualRate":12} Steps: Solve for total monthly installments n and convert to years and remaining months	{"totalMonths":161,"years":13,"remMonths":5}	{"totalMonths":161,"years":13,"remMonths":5}	**PASS**	tests/test_runner.js (TENURE-001)
TENURE-002	Zero interest rate: n = Target / P	In: {"targetCorpus":5000000,"monthlyAmount":50000,"annualRate":0} Steps: Pass 0% annual return rate; expect simple linear tenure calculation	{"totalMonths":100,"years":8,"remMonths":4,"totalGain":0}	{"totalMonths":100,"years":8,"remMonths":4,"totalGain":0}	**PASS**	tests/test_runner.js (TENURE-002)
TENURE-003	Zero target corpus input boundary	In: {"targetCorpus":0,"monthlyAmount":25000,"annualRate":12} Steps: Pass 0 target corpus and verify 0 months returned	{"totalMonths":0,"years":0}	{"totalMonths":0,"years":0}	**PASS**	tests/test_runner.js (TENURE-003)
TENURE-004	Target corpus less than first monthly installment	In: {"targetCorpus":10000,"monthlyAmount":20000,"annualRate":12} Steps: Set Target = 10,000, Monthly Installment = 20,000	{"totalMonths":1}	{"totalMonths":1}	**PASS**	tests/test_runner.js (TENURE-004)
EDU-001	10-Year college goal with 10% education inflation and 12% portfolio growth	In: {"childAge":8,"collegeAge":18,"costToday":2500000,"infRate":10,"growthRate":12} Steps: Compute future college cost at age 18 and derive required monthly SIP	{"yearsLeft":10,"futureCost":"6484356 ± 1.0","reqSIP":"27909 ± 1.0"}	{"yearsLeft":10,"futureCost":6484356,"reqSIP":27909}	**PASS**	tests/test_runner.js (EDU-001)
EDU-002	Zero education inflation assumption	In: {"childAge":5,"collegeAge":18,"costToday":3000000,"infRate":0,"growthRate":12} Steps: Verify Future Cost equals Cost Today when inflation is 0%	{"futureCost":3000000,"yearsLeft":13}	{"futureCost":3000000,"yearsLeft":13}	**PASS**	tests/test_runner.js (EDU-002)

ID	Scenario	Input / Steps	Expected	Actual	Status	Evidence
EDU-003	Immediate college entrance boundary (Child Age >= College Age)	In: {"childAge":18,"collegeAge":18,"costToday":2000000,"infRate":8,"growthRate":12} Steps: Pass Child Age 18 and College Age 18; verify minimum 1 year horizon clamped to prevent division by zero	{"yearsLeft":1}	{"yearsLeft":1}	**PASS**	tests/test_runner.js (EDU-003)
MAR-001	7-Year wedding goal with 6% inflation and 12% investment return	In: {"years":7,"costToday":3000000,"infRate":6,"growthRate":12} Steps: Compute future wedding budget and solve for required monthly SIP over 84 months	{"futureCost":"4510891 ± 1.0","reqSIP":"34179 ± 1.0"}	{"futureCost":4510891,"reqSIP":34179}	**PASS**	tests/test_runner.js (MAR-001)
MAR-002	1-Year short horizon boundary	In: {"years":1,"costToday":1500000,"infRate":5,"growthRate":10} Steps: Set years = 1; verify inflation scales for 1 year and 12 monthly installments calculated	{"futureCost":1575000,"yearsLeft":1}	{"futureCost":1575000,"yearsLeft":1}	**PASS**	tests/test_runner.js (MAR-002)
MAR-003	Zero current budget input	In: {"years":5,"costToday":0,"infRate":6,"growthRate":12} Steps: Pass 0 budget today and verify 0 required SIP	{"futureCost":0,"reqSIP":0}	{"futureCost":0,"reqSIP":0}	**PASS**	tests/test_runner.js (MAR-003)
RET-001	30-Year horizon retirement planning with Fisher real yield drawdown	In: {"currAge":30,"retAge":60,"monthlyExpenses":50000,"infRate":6,"preRate":12,"postRate":0.085,"lifespanPostRet":25} Steps: Adjust monthly expenses for 30 years of inflation, compute 25-year post-retirement annuity corpus, solve pre-ret SIP	{"futMonthlyExp":"287175 ± 1.0","corpusRequired":"64532060 ± 10.0","reqSIP":"18281 ± 5.0"}	{"futMonthlyExp":287175,"corpusRequired":64532060,"reqSIP":18281}	**PASS**	tests/test_runner.js (RET-001)
RET-002	Real rate safety floor (rReal >= 1.5%) when inflation exceeds post-ret yield	In: {"currAge":30,"retAge":60,"monthlyExpenses":50000,"infRate":9,"preRate":12,"postRate":0.085} Steps: Pass 9% inflation with 8.5% post-ret yield; verify real yield is floored at 1.5%	{"rReal":0.015}	{"rReal":0.015}	**PASS**	tests/test_runner.js (RET-002)
RET-003	1-Year immediate retirement boundary	In: {"currAge":59,"retAge":60,"monthlyExpenses":100000,"infRate":6,"preRate":12} Steps: Set currAge = 59, retAge = 60; verify 1-year accumulation schedule	{"yearsToRet":1}	{"yearsToRet":1}	**PASS**	tests/test_runner.js (RET-003)
RET-004	Zero living expenses input	In: {"currAge":35,"retAge":60,"monthlyExpenses":0,"infRate":6,"preRate":12} Steps: Pass 0 for living expenses; expect 0 corpus required and 0 SIP	{"corpusRequired":0,"reqSIP":0}	{"corpusRequired":0,"reqSIP":0}	**PASS**	tests/test_runner.js (RET-004)
RET-005	Zero pre-retirement investment return (0% pre-rate)	In: {"currAge":40,"retAge":60,"monthlyExpenses":50000,"infRate":6,"preRate":0} Steps: Pass 0% pre-rate; verify reqSIP = corpusRequired / totalMonths	{"reqSIP":150143}	{"reqSIP":150143}	**PASS**	tests/test_runner.js (RET-005)
EMI-001	Standard 20-Year ■ 50 Lakhs loan at 8.5% interest rate	In: {"principal":5000000,"annualRate":8.5,"tenureYears":20} Steps: Compute monthly reducing-balance EMI for 240 installments and total interest	{"monthlyEMI":"43391.16 ± 0.01","totalPayment":"10413879 ± 1.0","totalInterest":"5413879 ± 1.0"}	{"monthlyEMI":43391.16,"totalPayment":10413879,"totalInterest":5413879}	**PASS**	tests/test_runner.js (EMI-001)
EMI-002	Zero interest rate loan (0% interest)	In: {"principal":1200000,"annualRate":0,"tenureYears":5} Steps: Set rate = 0% for 5-year loan; expect EMI = Principal / Months	{"monthlyEMI":20000,"totalPayment":1200000,"totalInterest":0}	{"monthlyEMI":20000,"totalPayment":1200000,"totalInterest":0}	**PASS**	tests/test_runner.js (EMI-002)

ID	Scenario	Input / Steps	Expected	Actual	Status	Evidence
EMI-003	1-Year short-duration loan (12 installments)	In: {"principal":100000,"annualRate":12,"tenureYears":1} Steps: Pass ■ 1 Lakh principal for 1 year at 12%	{"monthlyEMI":"8884.88 ± 0.05"}	{"monthlyEMI":8884.88}	**PASS**	tests/test_runner.js (EMI-003)
EMI-004	Zero loan principal input	In: {"principal":0,"annualRate":8.5,"tenureYears":15} Steps: Pass 0 principal; expect 0 monthly EMI	{"monthlyEMI":0,"totalPayment":0,"totalInterest":0}	{"monthlyEMI":0,"totalPayment":0,"totalInterest":0}	**PASS**	tests/test_runner.js (EMI-004)
EMI-005	Negative input sanitation	In: {"principal":-100000,"annualRate":-5,"tenureYears":-2} Steps: Pass negative values for loan principal, rate, and tenure	{"monthlyEMI":0,"totalInterest":0}	{"monthlyEMI":0,"totalInterest":0}	**PASS**	tests/test_runner.js (EMI-005)
ALLOC-001	Level 1 Conservative Mandate: ■ 1 Crore capital allocation	In: {"capital":10000000,"riskLevel":1} Steps: Verify percentage weightings and calculated rupee allocations sum to exactly 100% of capital	{"equity":1500000,"debt":6500000,"gold":1500000,"alt":500000,"total":10000000}	{"equity":1500000,"debt":6500000,"gold":1500000,"alt":500000,"total":10000000}	**PASS**	tests/test_runner.js (ALLOC-001)
ALLOC-002	Level 3 Balanced Growth Mandate: ■ 2.5 Crore capital allocation	In: {"capital":25000000,"riskLevel":3} Steps: Verify weights and absolute rupee values for ■ 2.5 Crore capital	{"equity":13750000,"debt":6250000,"gold":2500000,"alt":2500000}	{"equity":13750000,"debt":6250000,"gold":2500000,"alt":2500000}	**PASS**	tests/test_runner.js (ALLOC-002)
ALLOC-003	Level 5 High-Alpha Mandate: ■ 5 Crore capital allocation	In: {"capital":50000000,"riskLevel":5} Steps: Verify high-conviction alpha weightings on ■ 5 Crore capital	{"equity":40000000,"debt":2500000,"gold":2500000,"alt":5000000}	{"equity":40000000,"debt":2500000,"gold":2500000,"alt":5000000}	**PASS**	tests/test_runner.js (ALLOC-003)
API-001	Direct Plan – Growth scheme name regex matching	In: ["HDFC Flexi Cap Fund - Direct Plan - Growth","SBI Small Cap Fund - Direct Growth","Kotak Flexi Cap Fund - Growth - Direct","Parag Parikh Flexi Cap Fund - Direct Plan - Growth Option"] Steps: Pass 4 variations of Direct Growth scheme nomenclature to isDirectGrowth()	[true,true,true,true]	[true,true,true,true]	**PASS**	tests/test_runner.js (API-001)
API-002	Rejection of Regular and IDCW / Dividend schemes	In: ["ICICI Prudential Bluechip Fund - Regular Plan - Growth","HDFC Balanced Advantage Fund - Direct Plan - IDCW","Nippon India Small Cap Fund - Direct Plan - Dividend Reinvestment"] Steps: Pass Regular and IDCW scheme names to isDirectGrowth()	[false,false,false]	[false,false,false]	**PASS**	tests/test_runner.js (API-002)
API-003	Invalid scheme code raises typed MFDataError	In: {"schemeCode":9999999} Steps: Call fetchSchemeData(9999999); expect SCHEME_NOT_FOUND error	{"errorClass":"MFDataError","code":"SCHEME_NOT_FOUND"}	{"errorClass":"MFDataError","code":"SCHEME_NOT_FOUND"}	**PASS**	tests/test_runner.js (API-003)
API-004	Network abort / timeout handling via AbortController	In: {"signalAborted":true,"forceRefresh":true} Steps: Attempt API fetch with aborted signal; verify typed TIMEOUT/NETWORK_ERROR error is caught	{"errorClass":"MFDataError","code":"TIMEOUT or NETWORK_ERROR"}	{"errorClass":"MFDataError","code":"TIMEOUT"}	**PASS**	tests/test_runner.js (API-004)
API-005	Live AMFI scheme metadata and NAV history parsing	In: {"schemeCode":122639} Steps: Fetch scheme 122639; verify scheme metadata, latestNav > 0, and chronological history loaded	{"code":122639,"hasNav":true,"historyRecordsCount":"> 2500"}	{"code":122639,"latestNav":88.5547,"historyRecordsCount":3290}	**PASS**	tests/test_runner.js (API-005)

ID	Scenario	Input / Steps	Expected	Actual	Status	Evidence
REAL-001	Strict monotonic chronological NAV ordering (Oldest to Newest)	In: {"schemeCode":122639} Steps: Traverse sorted history array; confirm each date timestamp is strictly greater than previous	{"isMonotonic":true}	{"isMonotonic":true,"recordsCount":3290}	**PASS**	tests/test_runner.js (REAL-001)
REAL-002	Non-trading day forward resolution (Sunday rolls forward to Monday)	In: {"requestedDate":"2026-10-11"} Steps: Request NAV on Sunday 2026-10-11; verify resolution to Monday 2026-10-12 NAV	{"resolvedDate":"2026-10-12","nav":102.5,"offsetDays":1}	{"resolvedDate":"2026-10-12","nav":102.5,"offsetDays":1}	**PASS**	tests/test_runner.js (REAL-002)
REAL-003	7-Calendar-day limit guard on missing NAV forward search	In: {"targetDate":"2026-01-02","maxSearchDays":7} Steps: Request date where next available NAV is 13 days away; verify null returned	null	null	**PASS**	tests/test_runner.js (REAL-003)
REAL-004	Deterministic 36-month Real SIP backtest on controlled NAV fixture	In: {"monthlyAmount":10000,"years":3,"sipDay":1} Steps: Execute 3-year SIP (36 installments); verify installment count, total units, valuation, and gains	{"installmentsCount":36,"totalInvested":360000,"valuationGreaterThanInvested":true}	{"installmentsCount":36,"totalInvested":360000,"valuation":421010}	**PASS**	tests/test_runner.js (REAL-004)
REAL-005	Combined Lumpsum and monthly SIP backtest on historical NAV fixture	In: {"lumpsum":200000,"monthlyAmount":5000,"years":3} Steps: Compute initial lumpsum units + 36 SIP installments, sum total units, compute final portfolio value	{"totalInvested":380000,"totalUnitsEqualsSum":true}	{"totalInvested":380000,"lumpUnits":3927.73,"sipUnits":2775.3,"totalUnits":6703.03}	**PASS**	tests/test_runner.js (REAL-005)
REAL-006	3-Year Step-Up SIP (10% annual increment) schedule on Real NAV data	In: {"initialMonthly":10000,"stepUpPercent":10,"years":3} Steps: Execute 36 months of stepped contributions; verify annual schedule and total invested	{"totalInvested":397200,"scheduleYears":3,"yr1":10000,"yr2":11000,"yr3":12100}	{"totalInvested":397200,"yr1":10000,"yr2":11000,"yr3":12100}	**PASS**	tests/test_runner.js (REAL-006)
REAL-007	XIRR Newton-Raphson convergence on known dated cash flow	In: [{"date":"2024-01-01","amount":-100000},{"date":"2025-01-01","amount":110000}] Steps: Pass 2 dated cash flows to calculateXIRR; verify convergence within 0.05% of 10.0%	{"xirrPercent":"10.00% ± 0.05%"}	{"xirrPercent":"9.97%"}	**PASS**	tests/test_runner.js (REAL-007)
REAL-008	2-Year Real STP: Debt source redemption and Equity target accumulation	In: {"initialCorpus":1000000,"monthlyTransfer":25000,"years":2} Steps: Redeem source units monthly and purchase target units on same date NAV; compute dual valuations	{"transfersCount":24,"totalTransferred":600000,"remainingSourceValuation":"> 0","targetValuation":"> 0"}	{"transfersCount":24,"totalTransferred":600000,"totalWealth":1191598}	**PASS**	tests/test_runner.js (REAL-008)
REAL-009	Real SWP: Sustainable monthly payouts vs capital exhaustion guard	In: {"sustainableWithdrawal":10000,"excessiveWithdrawal":60000} Steps: Test normal SWP (24 payouts) and excessive withdrawal SWP; verify depletion triggers at month 2	{"normalExhausted":null,"normalPayouts":24,"excessiveExhaustedMonth":2,"excessiveRemainingUnits":0}	{"normalExhausted":null,"normalPayouts":24,"excessiveExhaustedMonth":2,"excessiveRemainingUnits":0}	**PASS**	tests/test_runner.js (REAL-009)
REAL-010	Insufficient history raises typed INSUFFICIENT_HISTORY error in strict mode	In: {"requestedYears":5,"availableYears":1,"allowPartial":false} Steps: Request 60-installment SIP on 1-year history fund; expect INSUFFICIENT_HISTORY exception	{"errorClass":"MFDataError","code":"INSUFFICIENT_HISTORY"}	{"errorClass":"MFDataError","code":"INSUFFICIENT_HISTORY"}	**PASS**	tests/test_runner.js (REAL-010)
REAL-011	Graceful partial history execution with transparent advisory notice	In: {"requestedYears":5,"allowPartial":true} Steps: Run SIP on incomplete dataset; verify isPartialHistory flag and descriptive user notice	{"isPartialHistory":true,"hasNotice":true,"targetInstallments":60}	{"isPartialHistory":true,"executedInstallments":12,"targetInstallments":60,"notice":"Fund Inception: 01-01-2025 (1 Yrs live data). Simulated across maximum available 12 installments."}	**PASS**	tests/test_runner.js (REAL-011)

ID	Scenario	Input / Steps	Expected	Actual	Status	Evidence
REAL-012	10-Year full historical SIP backtest on Parag Parikh Flexi Cap (Code 122639)	In: {"schemeCode":122639,"monthlyAmount":10000,"years":10,"sipDay":1} Steps: Execute 120 monthly installments of ■ 10,000 on live AMFI data; verify XIRR and wealth accumulation	{"installmentsCount":120,"totalInvested":1200000,"currentValuation":"> 25.0 Lakhs","xirrPercent":"> 14.0%"}	{"installmentsCount":120,"totalInvested":1200000,"valuation":2803910,"xirr":"16.33%"}	**PASS**	tests/test_runner.js (REAL-012)
MODE-001	Preservation of monthly investment and horizon inputs across mode toggles	In: {"monthlyAmount":25000,"years":10} Steps: Set monthly amount ■ 25,000 and 10-year horizon; switch modes and verify input parity	{"totalInvested":3000000,"inputsPreserved":true}	{"totalInvested":3000000,"inputsPreserved":true}	**PASS**	tests/test_runner.js (MODE-001)
MODE-002	UI controls dynamically adapt between Real and Hypothetical modes	In: {"modesTested":["hypothetical","real"]} Steps: Verify rateGroup is shown only in Hypothetical mode, and scheme selectors only in Real mode	{"hypo":{"rateGroup":true,"realControls":false},"real":{"rateGroup":false,"realControls":true}}	{"hypo":{"rateGroup":true,"realControls":false},"real":{"rateGroup":false,"realControls":true}}	**PASS**	tests/test_runner.js (MODE-002)
MODE-003	Prohibition of silent fallback from Real NAV errors to CAGR projection	In: {"invalidScheme":8888888} Steps: Trigger real data failure; verify explicit MFDataError is raised instead of silent CAGR substitution	{"silentFallback":false,"raisesTypedError":true}	{"silentFallback":false,"raisesTypedError":true}	**PASS**	tests/test_runner.js (MODE-003)
MODE-004	User-directed error recovery button provides seamless switch to Hypothetical mode	In: {"errorBannerContent":"\n \n Live Data Feed Error\n Switch to Hypothetical CAGR Projection\n \n "} Steps: Verify error banner embeds an accessible button to switch mode on user intent	{"hasSwitchButton":true}	{"hasSwitchButton":true}	**PASS**	tests/test_runner.js (MODE-004)

7. Defect Log and Retesting Results

All defects observed during previous testing cycles have been addressed and validated:

Defect ID	Module	Severity	Description & Root Cause	Status	Retest Result
DEF-001	Calculators Layout	High	Class collision on '.hero' selector in calculators.html caused modal/drawer container to inherit 700px desktop height and distort calculator forms.	**Closed / Fixed**	PASS — Isolated selector to '.calculator-hero-container'.
DEF-002	Planner Cards	Medium	Inconsistent card heights and misaligned action buttons across Goal Planners on 1366x768 viewports.	**Closed / Fixed**	PASS — Grid display with equal-height flex column containers applied.
DEF-003	SIP Range Slider	Medium	Tick mark labels on Monthly Investment slider overlapped ('■1,000' and '■25,000' collided due to narrow 9.6% positioning offset).	**Closed / Fixed**	PASS — Spaced tick distribution to [1k, 50k, 100k, 175k, 250k].
DEF-004	Historical NAV	Medium	Schemes with inception dates under 10 years caused undefined reference errors when backtesting over full 10-year slider setting.	**Closed / Fixed**	PASS — Dynamic slider clamp and partial history alert banner implemented.

8. Mutual Fund NAV and API Validation

- **Live AMFI Integration:** Successfully validated live requests to MFAPI.in schema endpoints.
- **Debounced Search:** Autocomplete suggestions reliably isolate active schemes with high responsiveness.
- **Fault Tolerance:** Malformed payloads or HTTP timeouts trigger contextual error messages and allow fallback to mathematical mode.

9. Historical Calculation Validation

- **Real NAV Backtesting:** Accurate calculation of accumulated units across daily AMFI time-series.
- **Annualized Return (XIRR/CAGR):** Exact match with internal rate of return equations.
- **Dual-Mode Switcher:** Synchronous toggle between historical simulation and hypothetical CAGR without state corruption.

10. Evidence and Screenshots

- Test Runner Console Log: 'tests/test_runner.js' completed with 100% pass rate in 609ms.
- Structured Execution Dataset: 'tests/test_results.json' and 'qa-report/test_execution_results.csv'.
- Live Web Portal: Deployed and verified on Vercel at https://testing-phase1.vercel.app/calculators.

11. Conclusion

All 61 test cases have executed successfully with 100% pass rate and zero remaining critical defects. The application demonstrates institutional-grade mathematical accuracy and UI reliability. Certified ready for production.